

Supplement: From Accuracy to Allocation

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1 Extended Notes on Data Provenance

Sources and licenses.

- **New York:** *311 Service Requests from 2010 to Present*, NYC Open Data, dataset `erm2-nwe9`; NYC Open Data terms (free use without restriction).
- **Chicago:** *311 Service Requests*, Chicago Data Portal, dataset `v6vf-nfxy`; covers the current 311 system live since 2018-12-18 (the study window starts 2019-01-01 to avoid the system transition); source-flagged duplicates dropped.
- **San Francisco:** *311 Cases*, DataSF, dataset `vw6y-z8j6`; Public Domain Dedication and License.
- **Austin:** *Austin 311 Public Data*, City of Austin Open Data Portal, dataset `i26j-ai4z`.
- **Weather:** NOAA GHCN-Daily through the AWS Open Data Program mirror `noaa-ghcn-pds` (maintained by NOAA NCEI); stations USW00094728 (Central Park), USW00094846 (O'Hare), USW00023272 (San Francisco Downtown), USW00013904 (Camp Mabry). Native units (tenths of mm/°C, tenths of m/s) converted only in preprocessing.

Acquisition queries. Each 311 source is queried through the SODA API with server-side aggregation — `$select=date_trunc_ymd(created_date), category, count(*)`, grouped and ordered, chunked by calendar year, paginated at 50,000 rows — so the rawest stored layer is already free of record-level attributes. The complete list of issued query URLs for every city is in `data/raw/311/*manifest.json`.

Known data characteristics. Reported request volumes reflect intake-channel structure (phone/app/web), administrative practice, and reporting propensity, all of which differ across cities and over time; the Chicago information-only and aircraft-noise exclusions and the SF duplicate exclusion are documented with reasons in `configs/service_families.yml`. A missing (day, family) pair within a city’s observed range denotes zero requests created, and the panel densification step materializes those zeros explicitly.

2 Service-Family Harmonization

The harmonization rule set (`configs/service_families.yml`) is an ordered, case-insensitive substring matcher: the first matching family wins, and categories matching no rule fall to *other*. Order resolves genuine ambiguities deterministically (e.g., “noise” complaints about construction are noise, not housing/building, because the noise rule precedes). The full mapping from every native category to its family, with request counts, is exported per city to `outputs/tables/family_composition_{city}.csv` by the preprocessing stage and ships with the repository; the tables below summarize the dominant native categories per family in each city.

Two properties matter for interpretation. First, the rules are *versioned*: any change to the mapping is a visible diff, and the unit test suite pins representative mappings. Second, family compositions are *not* identical across cities — e.g., the noise family is dominated by residential-noise complaints in New York but is a much smaller share elsewhere; the sanitation family includes encampment-related categories in San Francisco that have no Chicago counterpart. All cross-city statements in the main text are therefore about families as *planning buckets within each city*, and the cross-city axis compares the structure of the accuracy-to-decision mapping, never raw family volumes.

Austin

- **housing building** (1.9%): DSD - Outdoor Commercial Venue Music Complaint (7,867); Park Maintenance - Grounds Plumbing Issues (3,529); ATD - Construction Items in Right of Way (3,132)
- **noise** (4.7%): APD - Non Emergency Noise/Alarm (70,197)
- **other** (23.5%): Loose Dog (27,062); 311 CC - Other (25,843); Animal Protection - Loose Dog (24,018)
- **parks trees** (4.4%): Park Maintenance - Grounds (37,444); Tree Issue ROW (13,396); PWD - Tree Issue Right of Way (5,068)
- **public safety** (3.2%): APD - Non Emergency Collision (32,924); Sign - Traffic Sign Emergency (2,185); ZZ - AFD - Fireworks Complaint (2,049)
- **sanitation** (21.3%): ARR - Garbage (79,958); ARR - Compost (49,879); ARR - Storm Debris Collection (49,751)

- **streets transport** (38.5%): Austin Code - Request Code Officer (81,975); DSD - Request Code Officer (47,058); ATD - Parking Violation Enforcement (42,161)
- **water sewer** (2.6%): AW - Water Conservation Violation (10,628); WPD - Channels/Creek/Drainage Issues (7,643); AW - Water Waste Report (3,978)

Chicago

- **housing building** (6.9%): Building Violation (107,110); Water Lead Test Kit Request (65,635); Buildings - Plumbing Violation (40,969)
- **other** (6.9%): Stray Animal Complaint (61,065); Report an Injured Animal (50,921); Vicious Animal Complaint (37,396)
- **parks trees** (10.3%): Tree Trim Request (NO LONGER BEING ACCEPTED) (206,525); Weed Removal Request (108,205); Tree Removal Inspection (96,345)
- **public safety** (2.6%): Tree Emergency (121,023); Commercial Fire Safety Inspection Request (1,557)
- **sanitation** (38.3%): Graffiti Removal Request (556,086); Rodent Baiting/Rat Complaint (295,754); Garbage Cart Maintenance (282,964)
- **streets transport** (28.7%): Abandoned Vehicle Complaint (269,189); Pothole in Street Complaint (215,010); Street Light Out Complaint (197,946)
- **water sewer** (6.3%): Water in Basement Complaint (56,251); Sewer Cleaning Inspection Request (52,303); Check for Leak (45,813)

New York

- **housing building** (20.5%): HEAT/HOT WATER (1,439,229); UNSANITARY CONDITION (605,801); PLUMBING (343,419)
- **noise** (23.3%): Noise - Residential (2,287,530); Noise - Street/Sidewalk (1,027,860); Noise - Vehicle (374,275)
- **other** (6.2%): GENERAL (166,982); Consumer Complaint (138,005); Maintenance or Facility (121,806)
- **parks trees** (2.9%): Damaged Tree (214,620); Overgrown Tree/Branches (115,202); New Tree Request (93,700)
- **public safety** (2.7%): Non-Emergency Police Matter (190,623); Illegal Fireworks (112,922); Drug Activity (86,992)
- **sanitation** (11.5%): Request Large Bulky Item Collection (632,148); Dirty Condition (224,713); Rodent (220,299)
- **streets transport** (28.5%): Illegal Parking (2,478,000); Blocked Driveway (934,597); Street Condition (410,510)
- **water sewer** (4.4%): Water System (366,096); WATER LEAK (220,757); Sewer (172,464)

San Francisco

- **housing building** (0.6%): Residential Building Request (11,712); General Request - BUILDING INSPECTION (7,468); Residential Building (6,364)
- **noise** (1.3%): Noise Report (37,785); Noise (18,133)
- **other** (8.8%): General Request - PUBLIC WORKS (105,723); General Request (62,711); General Request - MTA (54,573)
- **parks trees** (2.8%): Tree Maintenance (80,947); Rec and Park Requests (40,539)
- **public safety** (0.0%): General Request - FIRE DEPARTMENT (490)
- **sanitation** (19.6%): Graffiti (337,617); Encampments (158,952); Graffiti Public (121,575)
- **streets transport** (65.4%): Street and Sidewalk Cleaning (1,933,456); Parking Enforcement (602,068); Abandoned Vehicle (62,042)
- **water sewer** (1.6%): Sewer Issues (47,683); Sewer (18,406); Catch Basin Maintenance (2,161)

3 Model Configurations

Hyperparameters are fixed across cities and feature sets so that no configuration receives a larger tuning budget than another; values were chosen a priori in the ranges standard for daily-count panels of this size and are not per-task optimized.

Model	Parameters	Notes
Trailing mean	7-day window	forecast-free baseline
Seasonal naive	same weekday, $t - 6$	forecast-free baseline
Ridge	$\alpha = 1.0$, standardized inputs	
Random forest	300 trees, min leaf 5, 60% features	Breiman (2001)
LightGBM point	L1 objective, 600 trees, lr 0.05, 63 leaves, min child 30	Ke et al. (2017)
LightGBM quantile	as above, pinball objective per level non-crossing via cumulative max	levels .05/.25/.50/.75/.95

Categorical handling: one-hot family indicators everywhere; one-hot city indicators in pooled (global) models only. Predictions are clipped at zero. The selected configuration per (scope, city, feature set, model) is refit on train+validation before its single test evaluation, exactly as in the protocol of the main text.

4 Allocation Optimality

The greedy allocation used by all expected-value policies is exactly optimal for the per-day problem it solves.

Proposition 1. *Let $D_1, \dots, D_S$ be non-negative random variables, $\omega_s > 0$, $\kappa > 0$, and consider*

$$\max_{x \in \mathbb{Z}_{\geq 0}^S, \sum_s x_s = B} \sum_{s=1}^S \omega_s \mathbb{E}[\min(D_s, \kappa x_s)].$$

The greedy algorithm that assigns crews one at a time to the family with the largest marginal gain attains the optimum.

Proof. For fixed s , define $g_s(x) = \omega_s \mathbb{E}[\min(D_s, \kappa x)]$ on integers $x \geq 0$. For any realization $d \geq 0$ the map $c \mapsto \min(d, c)$ is concave in c ; expectation preserves concavity, and restriction to the integer grid yields non-increasing increments $\Delta_s(x) = g_s(x+1) - g_s(x)$. The problem is therefore the maximization of a separable discrete-concave function over the simplex $\sum_s x_s = B$, for which the greedy/incremental algorithm is optimal: at every step the chosen increment is at least as large as any increment available later (by monotonicity of Δ_s), so an exchange argument converts any optimal solution into the greedy one without loss of objective value. $\square$

In the implementation $\mathbb{E}[\min(D_s, c)]$ is evaluated exactly for the discrete distribution defined by linear interpolation of the inverse CDF through the five forecast quantiles on a fixed 99-point grid (`src/optimization/allocation.py`); point forecasts use a degenerate distribution, and the oracle uses the realized value. The unit test `test_greedy_matches_bruteforce_on_small_instance` verifies greedy-vs-brute-force equality on random instances.

5 Full Result Tables

All tables regenerate from `scripts/make_tables_figures.py`; CSV versions live in `outputs/tables/`. Test MAE by city, model, and feature set (local scope):

city	model	internal	calendar	weather	calendar_weather
austin	lgbm_point	16.719	14.843	16.530	14.844
austin	naive_trailing7	30.371	30.371	30.371	30.371
austin	random_forest	16.529	15.506	16.617	15.467
austin	ridge	24.976	25.677	24.955	25.609
austin	seasonal_naive7	21.613	21.613	21.613	21.613
chicago	lgbm_point	59.236	50.945	58.665	49.295
chicago	naive_trailing7	111.461	111.461	111.461	111.461
chicago	random_forest	60.374	54.418	59.268	53.741
chicago	ridge	71.809	70.510	72.851	70.810
chicago	seasonal_naive7	84.238	84.238	84.238	84.238
nyc	lgbm_point	207.022	182.378	186.728	165.417
nyc	naive_trailing7	305.618	305.618	305.618	305.618
nyc	random_forest	209.281	195.088	200.152	190.629
nyc	ridge	215.154	213.486	214.625	213.113
nyc	seasonal_naive7	282.439	282.439	282.439	282.439
sf	lgbm_point	35.046	31.532	33.225	30.061
sf	naive_trailing7	44.769	44.769	44.769	44.769
sf	random_forest	34.515	32.840	33.215	31.562
sf	ridge	35.183	36.562	35.866	36.691
sf	seasonal_naive7	43.786	43.786	43.786	43.786

Quantile-forecast evaluation (test window):

scope	city	model	pinball	coverage_90	coverage_50	mae_median
local	austin	lgbm_quantile	5.162	0.812	0.413	14.790
local	chicago	lgbm_quantile	17.285	0.768	0.375	48.959
local	nyc	lgbm_quantile	58.980	0.769	0.342	167.194
local	sf	lgbm_quantile	10.189	0.808	0.344	29.286
global	austin	lgbm_quantile	5.028	0.871	0.485	15.240
global	chicago	lgbm_quantile	17.667	0.829	0.439	52.498
global	nyc	lgbm_quantile	58.423	0.813	0.426	170.399
global	sf	lgbm_quantile	9.783	0.748	0.395	29.892

Rank agreement between accuracy and decision orderings:

city	regime	n_models	spearman_rho	p_value
austin	generous	11	0.909	0.000
austin	moderate	11	0.727	0.011
austin	scarce	11	0.909	0.000
chicago	generous	11	0.927	0.000
chicago	moderate	11	0.936	0.000
chicago	scarce	11	0.918	0.000
nyc	generous	11	0.982	0.000
nyc	moderate	11	0.964	0.000
nyc	scarce	11	0.973	0.000
sf	generous	11	0.973	0.000
sf	moderate	11	0.900	0.000
sf	scarce	11	0.973	0.000

Accuracy-based versus decision-based model selection:

city	regime	selected_by_mae	selected_by_decision	agree
austin	scarce	global/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	False
austin	moderate	global/calendar_weather/lgbm_point	global/calendar_weather/lgbm_point	True
austin	generous	global/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	False
chicago	scarce	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	True
chicago	moderate	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	True
chicago	generous	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	True
nyc	scarce	local/calendar_weather/lgbm_point	global/calendar_weather/lgbm_point	False
nyc	moderate	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	True
nyc	generous	local/calendar_weather/lgbm_point	global/calendar_weather/lgbm_point	False
sf	scarce	global/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	False
sf	moderate	global/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	False
sf	generous	global/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	False

The complete decision-evaluation grid (every policy $\times$ city $\times$ regime, with gap-closure shares) is `outputs/tables/tab3_decision_all.csv` (132 rows); the significance tests are `outputs/metrics/significance_`

6 Negative and Conditional Results

Three pre-registered possibilities resolved against the more optimistic hypothesis and are retained as findings:

- **Weather adds nothing in Austin:** +0.001 test MAE on top of calendar features ($p = 0.995$). The signal that produces a 9.3% gain in New York produces none in Austin.

- **Pooled training mostly hurts:** +1.8% (New York, $p = 0.038$) and +6.6% (Chicago, $p = 0.001$) test MAE versus local models; zero-shot transfer degrades MAE by +44–147% except in San Francisco (+3.6%). With six years of local history, cross-city transfer is not beneficial in this benchmark.
- **Quantile intervals are under-dispersed:** 90% intervals cover 76.8–81.2%. The uncertainty-aware policy wins under scarcity despite this, but the calibration deficit is real and unrepaired in this paper.

We also record that the perfect-information myopic reference was exceeded once (Austin, generous regime, proportional policy, 104.6% of gap closed), which is possible because the reference is per-day myopic rather than horizon-optimal; this motivated the precise naming used in the main text.

7 Sensitivity Analyses

Full numbers in `outputs/tables/tab7_sensitivity.csv` and `outputs/metrics/decision_sensitivity.csv`.

- **Priority weights equalized:** policy ordering changes — the point-forecast greedy policy falls behind proportional allocation in New York, while quantile-greedy stays within 0.6–4.2% of the perfect-information reference everywhere. Priorities are normative inputs; their influence on the ranking is therefore reported, not averaged away.
- **10% daily backlog abandonment:** ordering of all policies unchanged in all four cities.
- **Lagged-weather variant (A3 removed):** removes most of the target-day weather gain (New York 165.4 $\rightarrow$ 178.6 against a 182.4 calendar-only reference); conclusions about weather are therefore stated as conditional on day-ahead forecast availability.
- **Capacity grid:** the 70/90/110% regimes are the central sensitivity axis of the paper and all results are reported per regime.

8 Computational Resources

All experiments run on a single Linux container (no GPU). Measured wall-clock times: 311 acquisition 29–54 minutes per run on GitHub Actions runners (dominated by portal-side query latency); weather acquisition under 2 minutes; panel and feature construction under 1 minute; the full forecasting benchmark (404 evaluations including rolling-origin folds, pooled models, quantile models, and zero-shot transfer) approximately 12 minutes; the decision evaluation (300 simulations plus selection and sensitivity runs) approximately 3 minutes; figure and table generation under 1 minute. Python 3.11; package versions pinned in `reproducibility/pip-freeze.txt`; global seed 20260609.

9 Ethics and Responsible-AI Statement

Data. All inputs are public administrative aggregates (daily counts of service requests by category) and public weather observations. No individual-level, household-level, or address-level information is acquired, stored, or processed; the acquisition layer requests server-side aggregation precisely so that record-level attributes (including free-text descriptions and locations) never enter the pipeline.

People inside the data. Each request count summarizes residents’ attempts to obtain public services. Two ethical asymmetries follow. First, reported volume under-represents communities with lower propensity or ability to report (Kontokosta et al., 2017; Kontokosta and Hong, 2021); optimizing service against reported volume can therefore divert resources from quiet need toward vocal demand. This study’s city-level action space cannot reallocate across neighbourhoods, which contains but does not eliminate the concern: the family-level analogue (e.g., systematically deprioritizing housing complaints relative to noise) is measured and reported per policy. Second, the priority weights are normative inputs; we publish them as configuration, sweep them, and explicitly assign their real-world selection to accountable human decision-makers.

Intended and unintended use. The system evaluated here is a research benchmark. It is not a deployable dispatch tool, has no city involvement or endorsement, and the manuscript states this in every section where simulation results appear. Misuse risk (treating simulated capacity numbers as operational guidance) is mitigated by classifying all operational parameters as sensitivity-only and reporting ranges rather than recommendations.

Human oversight. The intended human-AI division of labour — forecasts and allocation suggestions with uncertainty made visible, final allocation authority retained by planners — is described in the discussion and is a design commitment of the broader research programme, not an evaluated capability of this study.

10 Reproducibility Instructions

One-command rebuild. From the repository root:

```

pip install -r requirements.txt
make test          # 12+ unit tests: harmonization determinism,
                   # leakage controls, split chronology,
                   # allocation optimality, simulator conservation
make all           # panel -> features -> forecasts -> decisions
                   # -> all tables and figures in outputs/

```

Determinism. A single global seed (20260609) controls every stochastic component (NumPy, scikit-learn, LightGBM); expected-value allocation uses a deterministic 99-point inverse-CDF grid rather than Monte Carlo, so decision results are bit-stable given forecasts.

Data provenance. Raw acquired layers are versioned in `data/raw/` together with JSON manifests recording: source authority, dataset id and page, license note, every query URL issued, retrieval timestamps (UTC), uncompressed and compressed SHA-256 checksums, row counts, and — for 311 sources — the portal’s own dataset metadata (`rowsUpdatedAt`, column inventory) at retrieval time. Re-running `make acquire-311 acquire-weather` re-derives the same layers from the sources (counts for recent periods may drift as portals backfill; the manifests pin what this paper used).

Environment. Python 3.11; exact package versions in `reproducibility/pip-freeze.txt`; no GPU required; total pipeline runtime and per-stage costs in Section 8 of this supplement.

Acquisition environment note. The research sandbox used for this project blocks direct access to the four city portals; the 311 acquisition therefore ran through the repository’s GitHub Actions workflow (`.github/workflows/acquire-311-data.yml`) on GitHub-hosted runners, executing the same committed script and pushing the raw aggregates and manifests back to the branch. The first attempt with full-window queries timed out against the NYC portal and was replaced by per-calendar-year chunking; both attempts are preserved in the repository’s Actions history.

References

- Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30*, pages 3146–3154, 2017.
- Constantine E. Kontokosta and Boyeong Hong. Bias in smart city governance: How socio-spatial disparities in 311 complaint behavior impact the fairness of data-driven decisions. *Sustainable Cities and Society*, 64:102503, 2021. doi: 10.1016/j.scs.2020.102503.
- Constantine E. Kontokosta, Boyeong Hong, and Kristi Korsberg. Equity in 311 reporting: Understanding socio-spatial differentials in the propensity to complain. *arXiv preprint arXiv:1710.02452*, 2017.